iSeq 100 Loading

A step-by-step guide

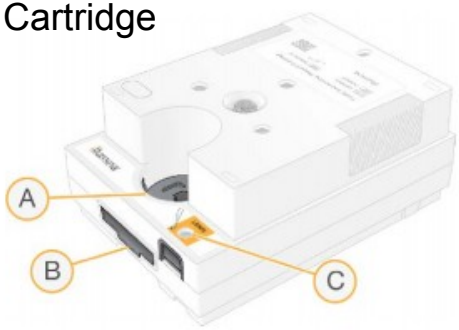


Rapid Response – Updated Jan 2026

Before Loading...

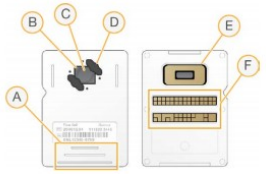
- Thaw Reagent Cartridge
  - Overnight in 4°C or
  - 6 hours in room temperature water bath
- Take out flow cell from 4◦C
  - At least 30 minutes before loading
- Dilute library to loading concentration (75pM or optimized concentrations)
  - Spike in 1% PhiX (default)
- Invert the cartridge 5 times then tap gently on table
- Insert flow cell into slot (B) in cartridge
- Carefully pierce foil with a pipette tip. Do not touch bottom of loading well
- Load 20µL of library into cartridge (C)

Cartridge



A Access window
B Flow cell slot
C Library reservoir

Flow cell



A Grip points
B CMOS sensor (top)
C Imaging area
D Gasket (one of two)
E CMOS sensor (bottom)
F Electrical interface

FS10000331 (0)



FS10000331 (0)

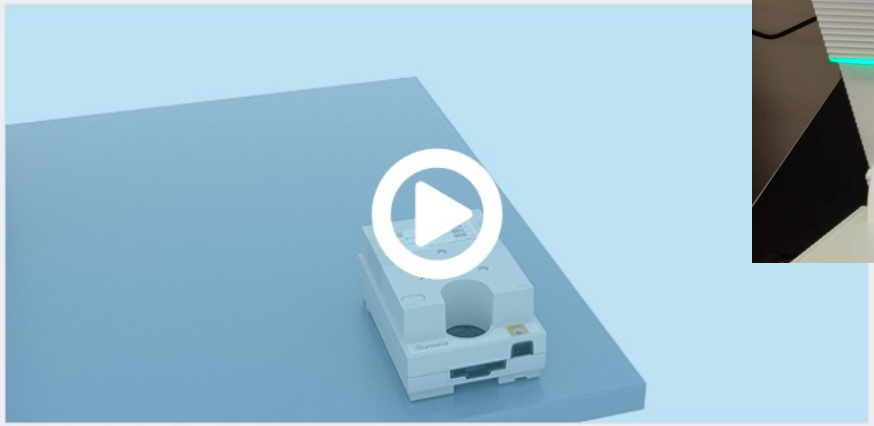
Opening Door ...



FS10000331 (0)

Load Cartridge

Insert the cartridge, and then select Close Door.
Play the video for preparation instructions.



Cancel Run

Close Door



FS10000331 (0)

Consumables Information

Flow Cell SN
BPA73104-1724

Lot Number
20410632



Cartridge SN
FY0026908-FC6

Lot Number
20402737

Run Cycles
322

Alerts (0)

Cancel Run

BaseSpace SEQUENCE HUB

michelle.tan@czbiohub.org

••••••••••

Sign In

Configuration

Run Monitoring Only

Run Monitoring and Storage

Skip BaseSpace Sequence Hub Sign In

FS10000331 (0)

Consumables Information

Flow Cell SN
BPA73104-1724

Lot Number
20410632



Cartridge SN
FY0026908-FC6

Lot Number
20402737

Run Cycles
322

Alerts (0)

Cancel Run

Run Setup

Run Name: Enter run name

Read Type: Single Read / Paired End

Read Cycle	Read 1	Index 1	Index 2	Read 2
	151	8	8	151

Advanced

Start Run

FS10000331 (0)

Consumables Information

Flow Cell SN
BPA73104-1724

Lot Number
20410632



Cartridge SN
FY0026908-FC6

Lot Number
20402737

Run Cycles
322

Alerts (0)

Cancel Run

Run Setup

Run Name: 20200122_Amy_K

Read Type: Single Read / Paired End

Read Cycle	Read 1	Index 1	Index 2	Read 2
	146	12	12	146

Advanced

Start Run

FS10000331 (0)

Consumables Information

Flow Cell SN
BPA73104-1724

Lot Number
20410632



Cartridge SN
FY0026908-FC6

Lot Number
20402737

Run Cycles
322

Alerts (0)

Cancel Run

Run Setup

Run Name: 20200122_Amy_K

Read Type: ○ Single Read ◉ Paired End

Read Cycle	Read 1	Index 1	Index 2	Read 2
	146	12	12	146

Advanced

Output Folder: G:\iSeq100-01 Browse

Sample Sheet: C:\Illumina\SampleSheet Browse

Start Run

FS10000331 (0)

Refer to training presentation for samplesheet instructions.

Consumables Information

Flow Cell SN
BPA73104-1724

Lot Number
20410632



Cartridge SN
FY0026908-FC6

Lot Number
20402737

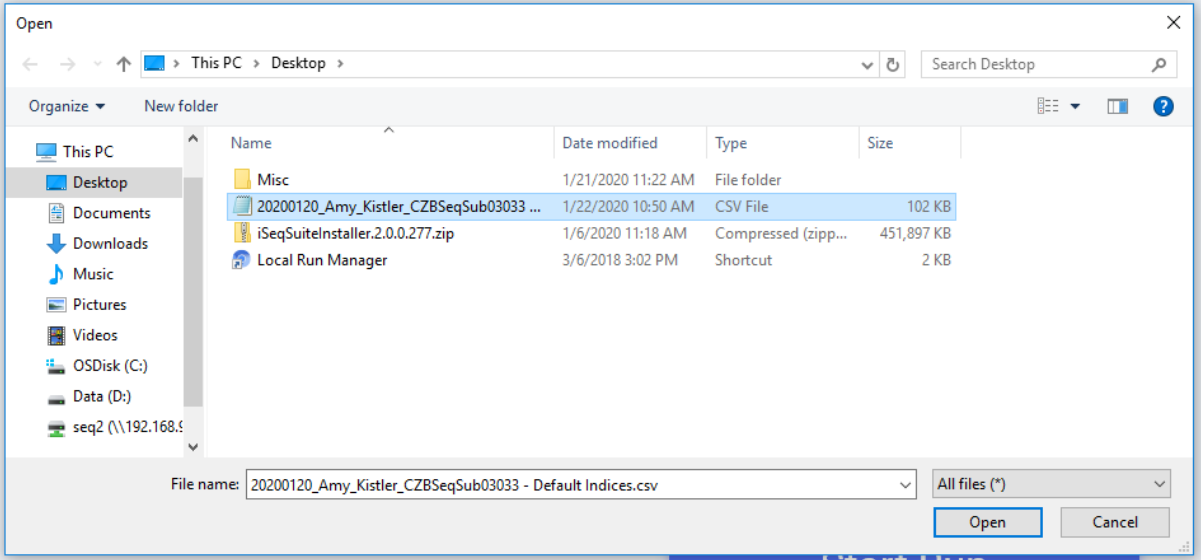
Run Cycles
322

Alerts (0)

Cancel Run

Run Setup

Run Name 20200122_Amy_K



Start Run

FS10000331 (0)

Consumables Information

Flow Cell SN
BPA73104-1724

Lot Number
20410632



Cartridge SN
FY0026908-FC6

Lot Number
20402737

Run Cycles
322

Alerts (0)

Cancel Run

Run Setup

Run Name: 20200122_Amy_K

Read Type: Single Read / Paired End

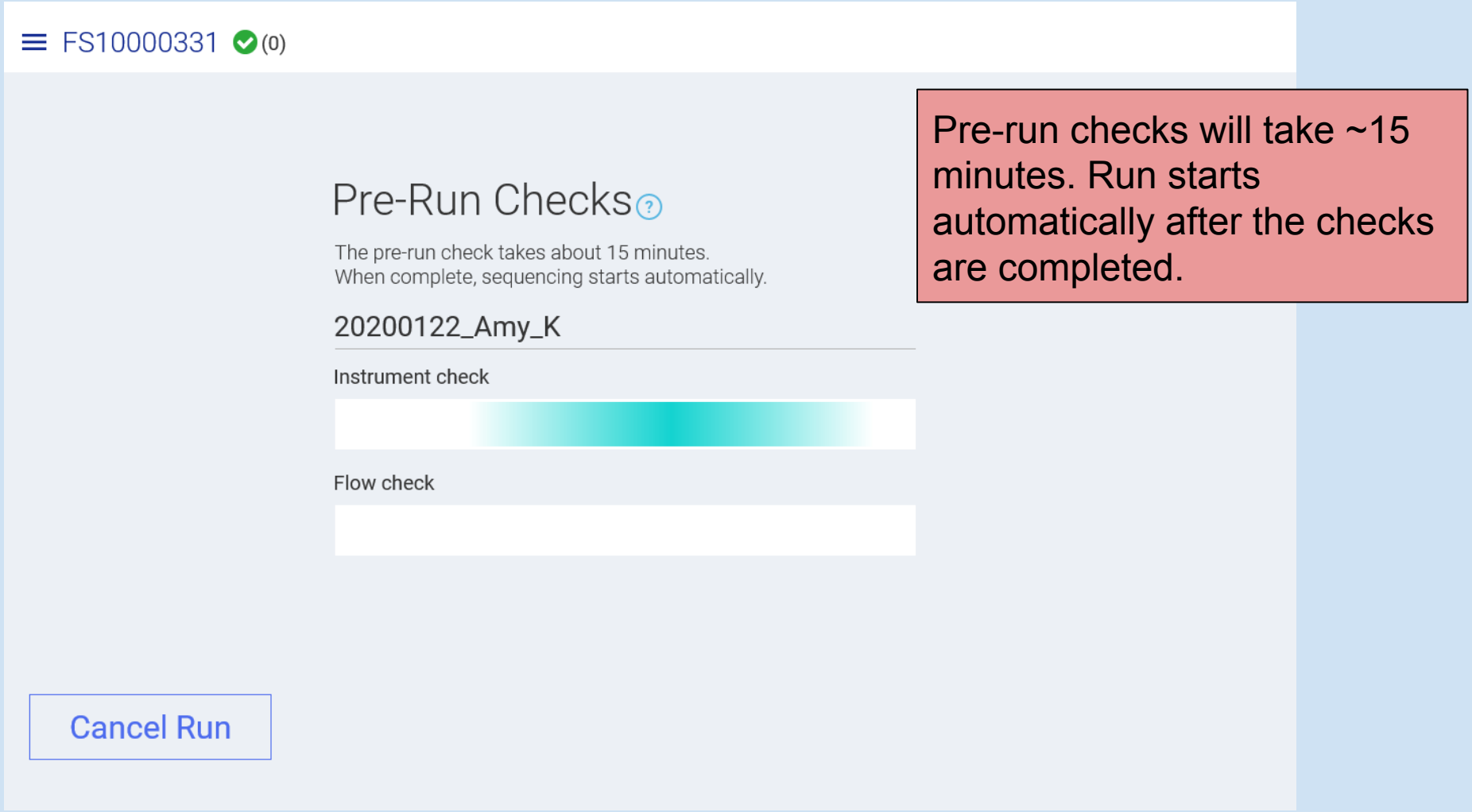
Read Cycle	Read 1	Index 1	Index 2	Read 2
	146	12	12	146

Advanced

Output Folder: G:\iSeq100-01 Browse

Sample Sheet: C:\Users\sbsuser.CZBIOH Browse

Start Run


FS10000331 (0)
Pre-Run Checks
The pre-run check takes about 15 minutes.
When complete, sequencing starts automatically.
20200122_Amy_K
Instrument check
Flow check
Cancel Run
Pre-run checks will take ~15 minutes. Run starts automatically after the checks are completed.

iSeq Settings

FS10000331 (0)

iSeq settings and options



FS10000331 (0)

- About
- System Check
- System Settings
- Software Update
- Process Management
- Local Run Manager
- Exit Application
- Shut Down System



About

FS10000331 (0)

Setti

Cancel

Save

iSeq™ Control Software Version 1.4.1.1700



Instrument Serial Number	**FS10000331**
Computer Name	**ISEQ100-01**
Computer IP Address	**192.168.99.129**
iSeq Server Version	**1.4.1.654**
Recipe Fragment Version	**6.1.4**
Total Run Count	**90**

Contact Us

www.illumina.com

United States
+1.800.809.ILMN
+1.800.809.4566

Customer Service
customerservice@illumina.com

International
+1.858.202.ILMN
+1.858.202.4566

Illumina Technical Support
techsupport@illumina.com

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FS10000331 (0)

System Check

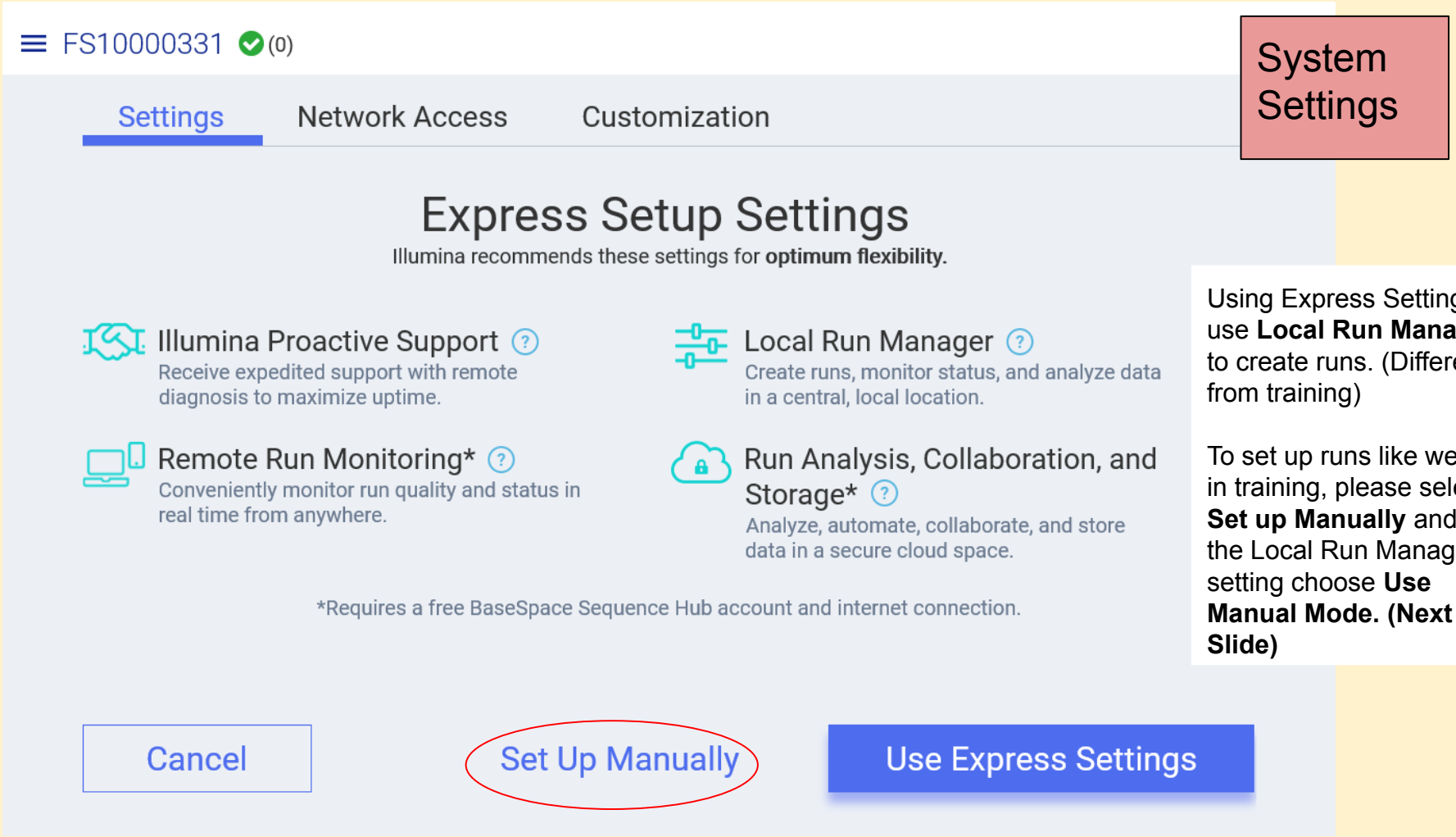
Do not remove the cartridge until the system check is complete. During mechanical test the software ejects and retracts the cartridge on



- [x] Mechanical Test
- [x] Thermal Test
- [x] Optical Test
- [x] Sensors Test

Unload | Load | Start

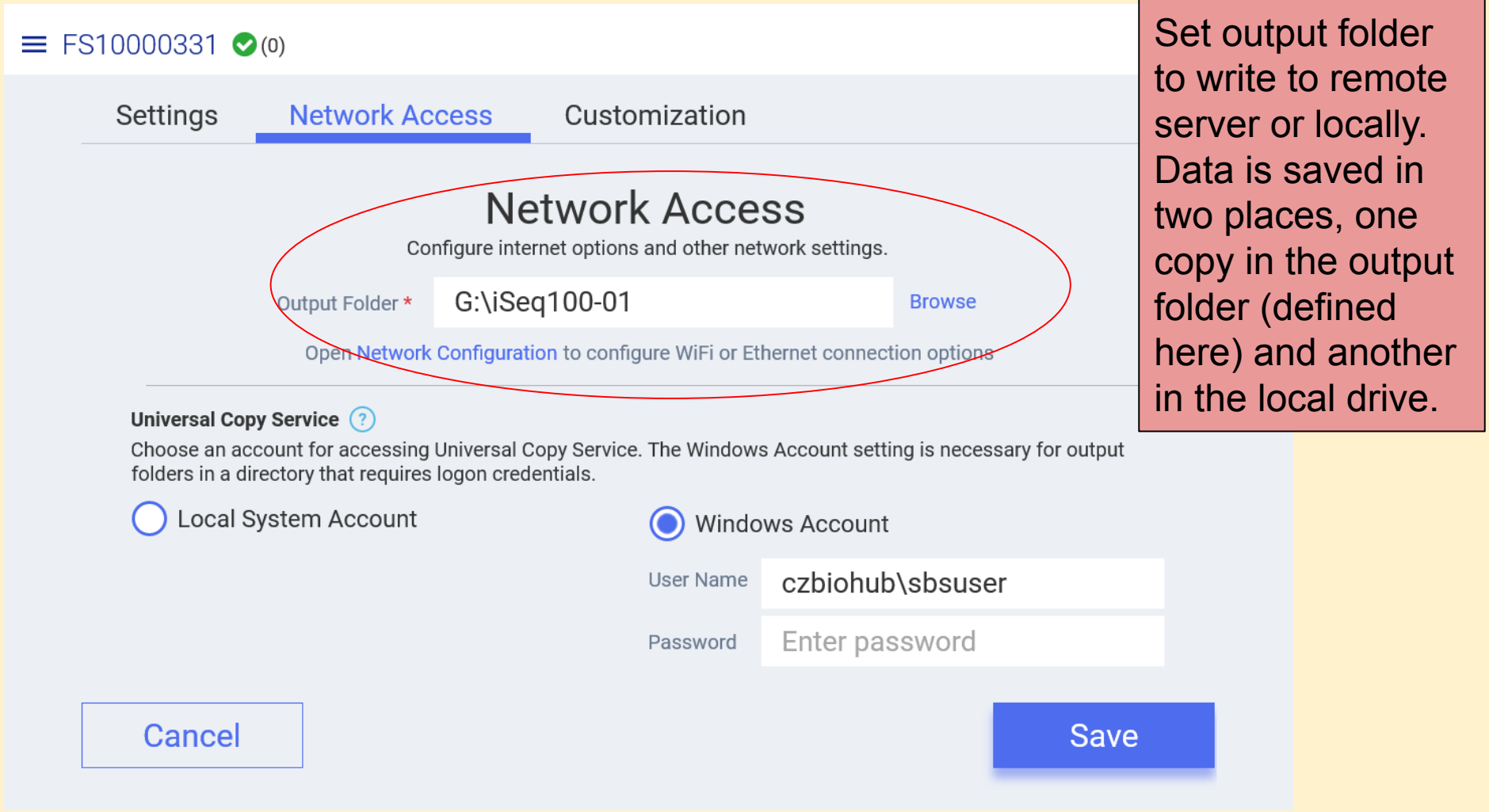
System Check: Remember to put in system test cartridge + flow cell.

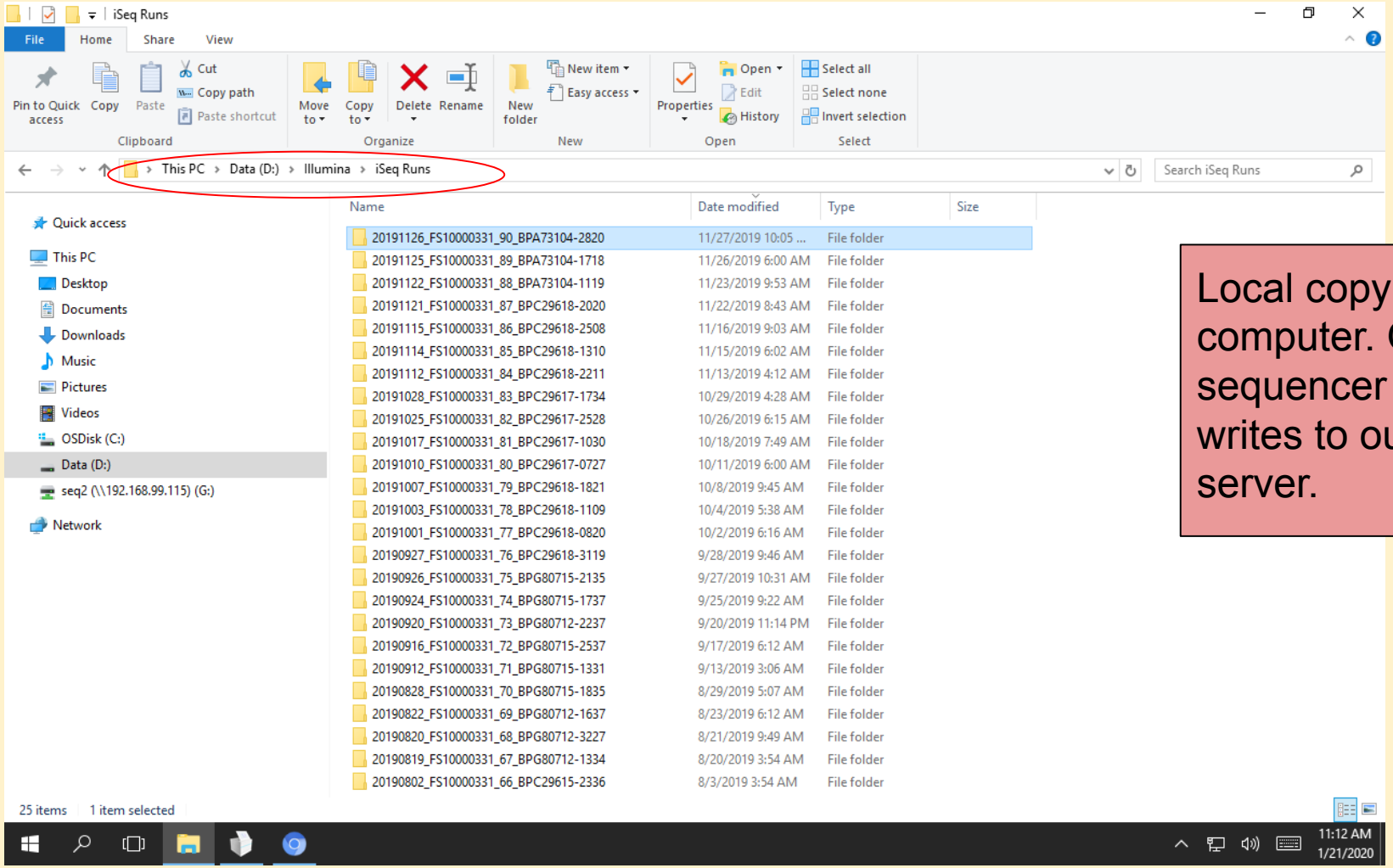


System Settings

Using Express Settings will use **Local Run Manager** to create runs. (Different from training)

To set up runs like we do in training, please select **Set up Manually** and in the Local Run Manager setting choose **Use Manual Mode. (Next Slide)**


FS10000331 (0)
Settings
Network Access
Customization
Network Access
Configure internet options and other network settings.
Output Folder *
G:\iSeq100-01
Browse
Open Network Configuration to configure WiFi or Ethernet connection options
Universal Copy Service
Choose an account for accessing Universal Copy Service. The Windows Account setting is necessary for output folders in a directory that requires logon credentials.
Local System Account
Windows Account
User Name
czbiohub\sbsuser
Password
Enter password
Cancel
Save
Set output folder to write to remote server or locally. Data is saved in two places, one copy in the output folder (defined here) and another in the local drive.



Local copy on computer. Our sequencer also writes to our remote server.

FS10000331 (0)

Settings | Network Access | Customization

Instrument Customization

Name your instrument, select where to store run files locally, and other custom settings.

Instrument Nickname: FS10000331

Audio: Off / On

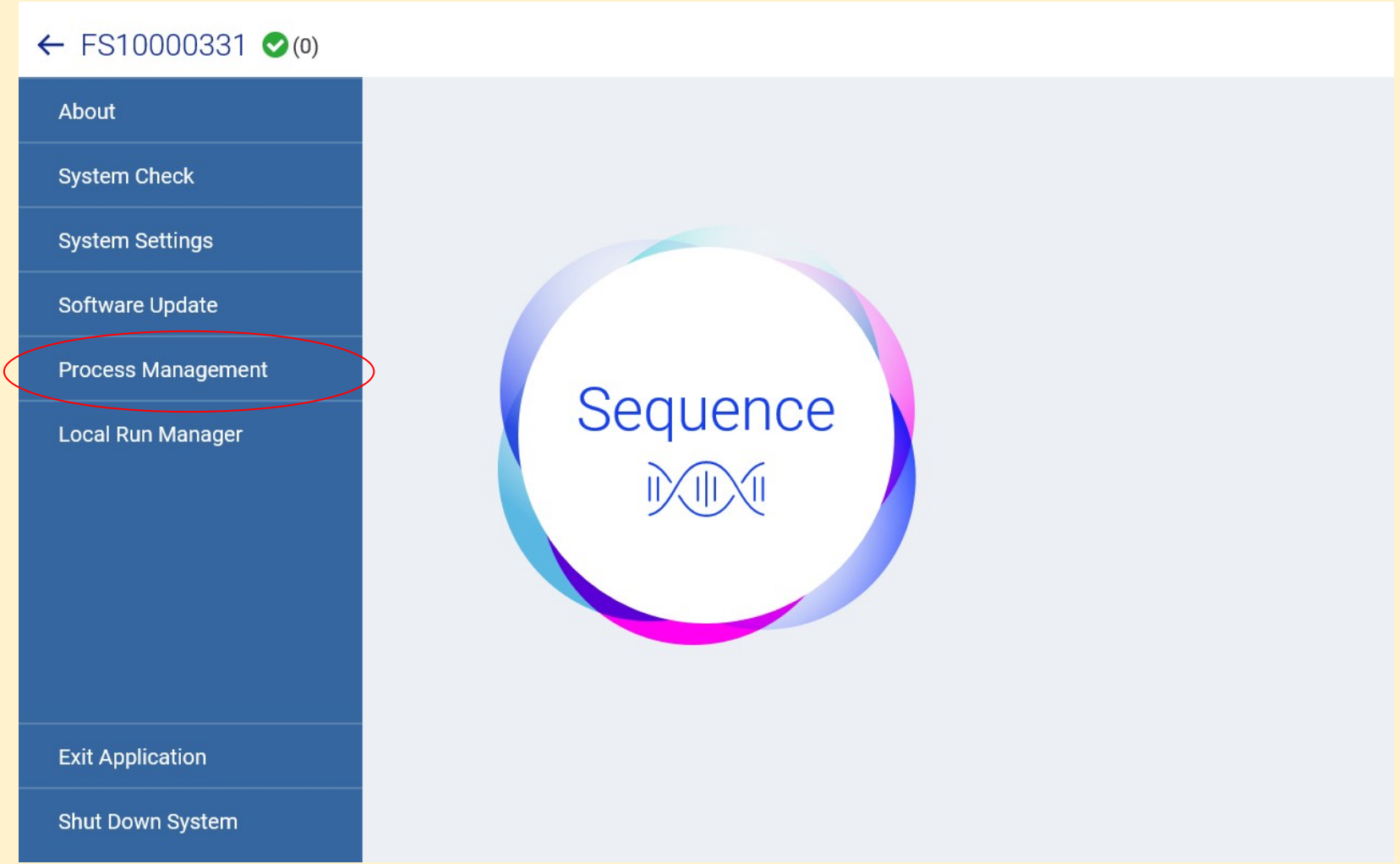
Save all thumbnail images

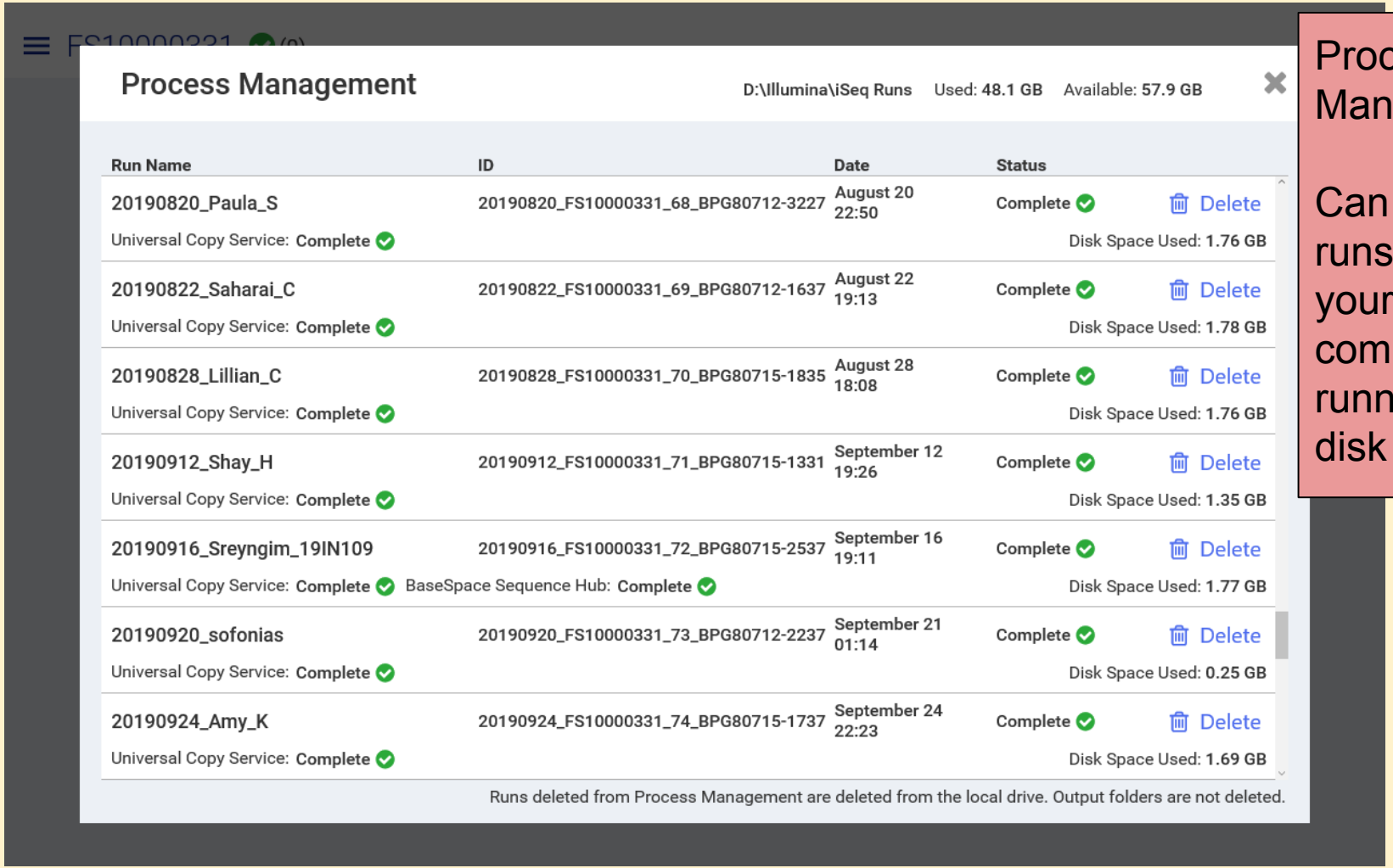
Software Updates: Autocheck for software updates

Requires an internet connection.

Cancel

Save


FS10000331 (0)
About
System Check
System Settings
Software Update
Process Management
Local Run Manager
Exit Application
Shut Down System
Sequence



Process Management:

Can delete runs here if your computer is running out of disk space

Next: 08-I_Sequencing Run QC for Illumina – Slides & Activity